

Abena Boadi-Agyemang

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 <https://aboadiag.github.io/>

 Pittsburgh, PA, USA

Research Interests

I am an interaction & design researcher who uses human-centered & participatory approaches to designing AI agents & robots that are **intuitive**, **safe**, and **inclusive** for *people with disabilities* in their daily lives.

Keywords – human-robot/AI interaction, human-computer interaction, human-centered design, accessibility, social robots, assistive agents, service robots.

Skills

robotics | programming: python, javascript (react.js; node.js), web APIs (REST, Fetch)
embedded hardware: mechanical design (Solidworks, Fusion 360), circuit design, embedded C
robot learning: imitation learning, reinforcement learning from human feedback (RLHF)

UX research | usability: A/B testing, think aloud, contextual inquiry
concept testing: WoZ methodology; wireframe- Figma, Adobe Xd; **analysis:** thematic, affinity diagramming; **tools** (Unity 3d, processing)

design thinking | participatory workshops: need-finding, ideation, prototyping;
multi-stakeholder: value-sensitive design; **inclusive design:** mixed dis/ability workshops

human research | experimental studies, statistical analysis (JMP, Matlab, R); qualitative analysis (ATLAS.ti); interview, focus group, ethnography, survey (Qualtrics, Prolific)

Education

Carnegie Mellon University | Ph.D. in Robotics, School of Computer Science

August 2021 - present, Expected June 2026, Pittsburgh, PA, USA

- **Focus:** Human-Robot Interaction, Accessibility, Design Research
- **Committee:** Dr. Aaron Steinfeld (Chair), Dr. Jean Oh, Dr. Patrick Carrington, Dr. Cynthia L. Bennett (Google)

Carnegie Mellon University | M.S. in Robotics, School of Computer Science

August 2021 - May 2025, Pittsburgh, PA, USA

- **Focus:** Human-Robot Interaction, Accessibility, Design Research
- **Committee:** Dr. Aaron Steinfeld (Chair), Dr. Henny Admoni, Dr. Nikolas Martelaro, Michelle Zhao (Ph.D. Candidate)
- **Thesis:** “*Simulated Encounters of the Third Kind: Scenario-Based Approach to Designing Guide Robots*”

Stanford University | B.S. in Mechanical Engineering, School of Engineering

September 2017 - June 2021, Stanford, CA, USA

- **Focus:** Dynamic Systems & Controls
- **Capstone:** “*Using a Vibrotactile Stimulation Device to Improve Knee Extension During Stance Phase in Children with Spastic Cerebral Palsy*” (Teammates: Perez, G. A., Mhatre, K., Wang, J.) [**Nominated for Hoefer Prize for Writing in the Major**]

Experience

Robotics Institute, Carnegie Mellon University | Graduate Research Assistant

Aug 2021 - present, Pittsburgh, PA, USA

- Designed, executed, & managed independent user-research focused on diverse, digital & physical interactions between potential end-users with disabilities & robots used for: (1) navigation assistance; (2) on-demand delivery; (3) mental wellbeing.
- Qualitative & quantitative methodologies: ○ Longitudinal design engagements (3-6 weeks): robot prototyping & diary studies; ○ Experimental studies: human beliefs, perceptions, & behavior in human-AI/robot interaction; ○ Decision-making algorithms for adaptation of embodied & digital AI systems using behavioral data: CV-based (**serengil DeepFace**); physiological sensing, & user activity.
- Synthesize qualitative & data-driven insights to generate recommendations focused on improved interaction & collaboration between humans & robots/AI [**see recent publications 1-4**].

SHAPE Lab, Stanford University | Undergraduate Research Assistant

Jun 2019 - Jun 2021, Stanford, CA, USA

- Supervised by Dr. Sean Follmer. Mentored by Drs. Alexa F. Siu & Elyse D.Z. Chase.
- Designed & fabricated haptic devices to make GUIs more accessible for blind & low vision (BLV) users & provide non-visual adapted materials for BLV students.

Autodesk | Fusion Catalyst Student Intern

Jul 2020 - Dec 2020, Stanford, CA, USA

- Secured & oversaw community partnerships between Autodesk & three Stanford student organizations. Provided free Fusion 360 (CAD & CAM software) training to students.

VMware Women's Leadership Innovation Lab | Undergraduate Research Assistant

Oct 2018 - Jun 2019, Stanford, CA, USA

- Supervised by Dr. Melissa Abad. Conducted qualitative analysis of the gendered-language used in fortune 500 company leadership statements to further examine their effects on the work environment.
- Produced database of commonly-used words (by frequency) to inform inclusivity interventions for fortune 500 partners.

Graduate School of Education | Research Assistant

Nov 2017 - Mar 2018, Stanford, CA, USA

- Supervised by Dr. John Willinsky. Inspected & catalogued 20% of journals (on Beall's list) that use open source publishing software, Open Journals System & are classified as "predatory" (i.e., exploitative academic publishing).
- Prepared contact database of journal board members to solicit participation in interventions that improve their journal integrity.

**Academic
Publications**

(* denotes equal contribution. [Google Scholar](#) for full papers.

- **[Under Review, Journal]** Carter, Lopez, Sun, Warriar, Wu, Ng, Wang, **Boadi-Agyemang**, Tomasic, Zimmerman, and Steinfeld. Designing Future Smartphone Apps for Navigation, Exploration, and Object Identification with Blind Users.
- 1. **[Workshop Paper]** **Boadi-Agyemang, A.** (2026). Robot Design at the Margins: The Lived Experience as a Participatory Approach to Co-Designing with Disabled Adults. HRI Pioneer. In *Companion at ACM/IEEE HRI*.
- 2. **[Conference]** **Boadi-Agyemang, A.**, Moharana, S., Bennett, C.L., Carter, E.J., Carrington, P., Steinfeld, A. (2026). Pluriversal Approach to Co-Designing Delivery Robots with People with Disabilities. *Just accepted at ACM/IEEE HRI*.
- 3. **[Journal]** Shih, K.*, **Boadi-Agyemang, A.***, Carter, E.J., Steinfeld, A. (2025). A Multi-Method Investigation of Guide Robot Characteristics for Blind and Low Vision Users. In **ACM THRI**.
- 4. **[Conference]** **Boadi-Agyemang, A.**, Schaldenbrand, P., Misra, V., Carter, E.J., Oh, J., Steinfeld, A. (2025). If I Move, Do You Move? Investigating the Role of Interpersonal Synchrony in Human-Robot Joint Painting. In **IEEE RO-MAN**.
- 5. **[Workshop Paper]** **Boadi-Agyemang, A.***, Moharana, S.* (2025). The Case for Em(body)ed Approaches to Disability-Centered Interaction Design. "Body Politics: Unpacking Tensions and Future Perspectives for Body-Centric Design Research in HCI" at **ACM CHI 2025**.
- 6. **[Short Paper]** **Boadi-Agyemang, A.*** and Park, M.* (2024). Designing Interactive Agents to Support Emotion Regulation in the Workplace through Guided Art-Making. In *Companion at 2024 ACM/IEEE HRI*.
- 7. **[Conference]** Han, V. Y., **Boadi-Agyemang, A.**, Lin, Y., Lindlbauer, D., and Ion, A. (2023). Parametric Haptics: Versatile Geometry-based Tactile Feedback Devices. **ACM UIST**.
- 8. **[Workshop Paper]** **Boadi-Agyemang, A.***, Shih, K.* and Steinfeld, A. (2023). Discovering User Needs and Preferences for Guide Robots: Challenges and Preliminary Insights. "Assistive Robots for Citizens" at **IEEE/RSJ IROS**.
- 9. **[Short Paper]** **Boadi-Agyemang, A.**, Carter, E.J., Siu, A. F., Steinfeld, A., and Martinez, M.O. (2023). Understanding Experiences, Attitudes, and Perspectives towards Designing Interactive Creative Tools for Teachers of Visually Impaired Students. In **ACM ASSETS**.
- 10. **[Book Chapter]** Siu, A. F., Chase, E. D. Z., Kim, G. S. H., **Boadi-Agyemang, A.**, Gonzalez, E. J., and Follmer, S. (2021). Haptic Guidance to Support Design Education and Collaboration for Blind and Visually Impaired People. **Springer**.

11. [Short Paper] Kim, L. H.*, **Boadi-Agyemang, A.***, Siu, A. F., and Tang, J. (2020). When to Add Human Narration to Photo- Sharing Social Media. *In ASSETS*.
12. [Short Paper] Chase, E. D. Z., Siu, A. F., **Boadi-Agyemang, A.**, Kim, G. S. H., Gonzalez, E. J., and Follmer, S. (2020). PantoGuide: A Haptic and Audio Guidance System to Support Tactile Graphics Exploration. *In ASSETS*.

Select Awards

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- HRI Pioneer '26. 1 of 22 selected from 96 applicants.
 - Carnegie Mellon University Uber Presidential PhD Fellowship ('22-'23).
 - Cadence Diversity in Technology Scholarship ('23): Students who exemplifies leadership & demonstrates a passion for technology.
 - University of Washington (UW) College of Engineering Dean's Fellowship [*declined*] ('21): For top incoming doctoral applicants (**\$10k/yr**).
 - UW Top Scholar Fellowship [*declined*], 2021: to attract outstanding students (**\$2.5k**).
 - Access_&Design Fellowship ('21): for emerging BIPOC designers.
 - Botswana Top Achievers Award and Scholarship ('17-'21): awarded by Botswana's Ministry of Education.
 - Stanford Mechanical Engineering Summer Undergraduate Research Institute (SURI) ('19, '20): fellowship to conduct a ten-week original research project (**\$7.5k**).
 - Academic Award of Excellence('18, '19): awarded by Stanford University's Black Community Services Center.